MIL-L-10028F
22 October 1985
SUPERSEDING
MIL-L-10028E
3 March 1981

MILITARY SPECIFICATION

LANYARD, INDIVIDUAL EQUIPMENT CARRYING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and size of pistol lanyard.

1.2 Classification. The pistol lanyard shall be of the following classes, as specified (see 6.2).

Class 1 - White
Class 2 - Olive Green 106

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-276	- Thread, Cotton
TT-E-529	- Enamel, Alkyd, Semigloss
PPP-B-566	- Boxes, Folding, Paperboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8465

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PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-676 - Boxes, Setup

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

2-4-56 - Lanyard, Pistol

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. Unless otherwise specified, the following document of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 633 - Electrodeposited Coatings of zinc on iron and steel
D 756 - Resistance of Plastics to Accelerated Service Conditions
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

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3. REQUIREMENTS

3.1 First article. When specified a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.5).

3.2 Materials (see 6.4).

3.2.1 Cotton yarn. The core yarn shall be made from cotton and shall be four tenacity, bright finish nylon.

3.2.2 Nylon yarn. The cover yarn shall be two or three ply, multifilament, high tenacity, bright finish nylon.

3.2.2.1 Color. The color of the nylon yarn for class 1 lanyard shall be white (undyed) and the color of the nylon yarn for class 2 lanyard shall be Olive Green 106 and shall match the standard sample (see 6.3). The dyed yarns for class 2 lanyards shall show good colorfastness to laundering and to light when tested as specified in 4.4.1.1.

3.2.2.2 Matching. The color and appearance of the dyed nylon yard shall match the standard sample when viewed under filtered tungstem lamps which approximate artificial daylight having a correlated color temperature of 7000 ± 500 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 100 K.

3.2.3 Thread, cotton. The thread shall conform to type IAL, 3 ply, ticket No. 30 of V-T-276.

3.2.4 Snaphook with swivel. The 1/2 inch snaphook shall be made of steel and shall conform to Waterbury Backly Company Part No. 402 or equal. The snaphook after being fully depressed, shall snap shut with no gap between the spring and the hook.

3.2.4.1 Finish. The finish for snaphook with swivel shall consist of a phosphate treated zinc plate, conforming to type IV, SC3 of ASTM B 633, followed by bake enameling. The enamel shall be black and shall conform to type II, class B or TT-E-529. The coating shall level out to a uniform continuous dry film of uniform black color and shall have no objectionable orange peel, wrinkles, drops, streaks, or areas of no film.

3.2.5 Staple. The wire for the staple shall be zinc or cadmium plated steel of 0.018 inch nominal thickness, and 0.043 inch nominal width. The crown width of the staple shall be 1/2 ± 1/32 inch and the leg length of the staple shall be 7/16 ± 1/32 inch.

3.2.6 Keeper, plastic. The top, bottom, and adjustable keepers shall be made of extruded vinyl-chloride-acetate copolymer plastic in a black color and in the thickness, shape, and size shown on Drawing 2-4-56. The keepers shall show no

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hardening, cracking, warping, or discoloration after being subjected to laundering, dry cleaning, and accelerated service (temperature and humidity extreme) when tested as specified in 4.4.1.1.

3.3 Construction. The construction shall conform in all respects to Drawing 2-4-56 and as specified herein.

3.3.1 Cords. The cords shall be braided with nylon yarn specified in 3.2.2, over a cotton core having a minimum of 5 ends of core yarn specified in 3.2.1. The cord shall be braided on a 16 carrier machine with a minimum of 2 ends per carrier, with carriers set in pairs, giving an interlacing of over-one and under-one basket stitch. The finished braided cord shall be free of grease or oil stains and shall conform to physical requirements specified in table I when tested as specified in 4.4.2.2.

TABLE I. Physical requirements of cord

Diameter (inches)1/	Breaking strength (pounds), minimum	Weight per unit length (ounces per liner yard), minimum	Picks per inch, minimum
3/16	225	0.30	9

1/ A tolerance of minus 1/32 inch is permitted.

3.3.1.1 Securing ends of cord. One end of the cord shall be seared to prevent ravelling and then stapled to the adjustable keeper with the end extending approximately 1/8 inch beyond the edge of the keeper. The other end of the cord shall be secured, after threading through the swivel eye, by hand tacking or machine stitching. Before hand tacking, the folded end of the cord shall be bound with not less than 10 wrappings of thread specified in 3.2.3. When hand tacking in a zig-zag pattern using stitch type 204 of FED-STD-751, not less than 30 stitches of the thread specified in 3.2.3. The bottom keeper shall cover the wrapping and stitching as a tight fit. The top keeper shall be free for adjusting purposes.

3.4 Workmanship. The finished pistol lanyard shall be clean and free from defects, such as cuts, frayed cord, cracked or damaged keepers, or imperfect snaphook.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Component testing. In addition to the quality assurance provisions of the subsidiary specifications and drawings, tests shall be performed on components and materials listed in table II for the characteristics shown. The sample unit shall be 5 yards of nylon yarn or three keepers, as applicable. (The last size shall be expressed in unit of cones or spools of yarn or in units of keepers, as applicable.) The inspection level shall be S-1 and the acceptable quality level (AGL), expressed in terms of defects per hundred units, shall be 6.5.

TABLE II. Component tests

Components and characteristics as	Requirement reference	Test method	No. of determinations per unit	Results reported
Dyed nylon yarn colorfastness to:				
laundering	3.2.2.1	5614 1/	1	Pass or fail
light	3.2.2.1	5660 1/	1	Pass or fail

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TABLE II. Component tests. (cont'd)

Components and Characteristics as	Requirement reference	Test method	No. of determinations per unit	Results reported
Keeper resistance to:				
laundering	3.2.6	5610 1/ 2/	1	Pass or fail
dry cleaning	3.2.6	5622 1/	1	Pass or fail
accerlerated service	3.2.6	D 756 3/	1	Pass or fail

1/ Method as specified in FED-STD-191

2/ Except that the soap solution shall not include sodium carbonate and acetric acid solution shall not be used.

3/ In accordance with one complete cycle of Procedure A, ASTM D 756.

4.4.2 In-process inspection.

4.4.2.1 In-process examination. Examination shall be made to determine that the following are as specified. Whenever nonconformance is noted, correction shall be made to the items affected and the process.

(a) Securing ends of cord for compliance with 3.3.1.1.
(b) End of cord in adjustable keeper is seared for compliance with 3.3.1.1.

4.4.2.2 In-process testing. The braided cord shall be tested for the characteristics listed in table III. The lot size shall be expressed units of spools or cones of cord. The sample unit shall be 5 yards of cord. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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TABLE III. In-process tests

Component and Characteristics	Requirement reference	Test method	Number of determinations per unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Cord:					
Diameter, inches	3.3.1	Std. Comm.	2	-	1/32 inch
Breaking strength, pounds	3.3.1	6016 1/	5	-	1 pound
Weight, ozs./liner yard	3.3.1	2/	1	-	0.01 ounce
Picks per inch	3.3.1	Visual	2	-	Number
Number of core yarns	3.3.1	Visual	1	x	-
Number of ends per carrier	3.3.1	Visual	1	x	-
Number of carriers	3.3.1	Visual	1	x	-
Braid	3.3.1	Visual	1	x	-

1/ Method as specified in FED-STD-191.

2/ A 48-inch test specimen of cord shall be suspended in such a manner as to allow a 2-ounce weight to be hung on the lower end and be free from abstruction. After 1 minute, two marks shall be made on the cord a distance of 36 inches apart. Release the tension, cut the specimen at the 36-inch gage marks and weigh to the nearest 0.01 ounce.

4.4.3 End item visual examination. The end item shall be examined for the defects listed in table IV. The lot size shall be expressed in units of lanyards. The sample unit shall be one completely fabricated lanyard. The inspection level shall II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 (for total major and minor combined) defects.

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TABLE IV. End item visual defects

Examine	Defect	Classification Major	Minor
Cord	Unevenly braided, i.e., not over-one and under-one basket stitch.		X
	Kinks, or unevenly braided resulting in open places, breaks in continuity of braid, or soft spots.	X	
	Broken, loose, or projecting strands.	X	
	End of cord extending beyond adjustable keeper not seared.		X
	Color and appearance not as specified.	X	
Plastic keepers	One or more missing.	X	
	Broken or damaged.	X	
	Any discoloration.		X
	Color other than black.	X	
	Misshapen or malformed, crack, puncture, or cut.	X	
Staple	Missing from adjustable keeper.	X	
	Broken.	X	
	Staple in adjustable keeper malformed.		X
	Improperly or insecurely clinched.		X
	Not attached as specified.		X
	Finish not as specified.		X
Snaphook with swivel	Component malformed, falls to function properly.	X	
	Spring does not close fully, causing gap between spring and hood.	X	
	Fractured, split, or broken.	X	
	Scratches, marks, or split.		X
	Any sharp edge or burr.	X	
	Finish peeling, blistered, or flaking.	X	
	Wrinkles, drops, streaks, or orange peel in finish.		X
	Not enameled.	X	
	Area of thin film or no film.		X
	Enamel not black.	X	

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Assembly	Component insecurely attached.	X	
	Any cut or tear.	X	
	Frayed.		X
	Bottom keeper not tight fit on wrapping; i.e., allowing keeper to slide off wrapping.		X
	Incorrectly assembled; e.g., arrangement not as specified.		X
	Adjustable keeper not operative.	X	
	Components not attached in location specified.		X
Cleanness	Noticeable grease or oil stains, thread ends not trimmed.		X

4.4.4 End item dimensional examination. The end item shall be examined or conformance to the dimensions specified on the drawing. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of lanyards. The sample unit shall be one lanyard. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 Packaging inspection. An inspection shall be made to determine that preservation, packing, and marking comply with section 5 requirements. Defects shall be scored in accordance with table V. The sample unit shall be one shipping container, fully packaged. The lot size shall be the number of shipping containers in the inspection lot. The inspection level shall be S-2 and the AQL expressed in terms of defects per hundred units, shall be 2.5.

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TABLE V. Packaging defects

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: open and non-continuous heat sealed seams and closure of polyethylene bags. Incorrectly fabricated bag. Incomplete closure of container flap, loose strapping or taping, or inadequate stapling. Bulged or distorted container.
Contents	Number per container is more or less than required.

4.4.6 Palletization inspection. An inspection shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 an AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2)

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5.1.1 Level A.

5.1.1.1 Unit packing Each lanyard shall be folded in half twice to a finished length of approximately 8 inches and inserted in a snug-fitting, flat style, clear polyethylene film bag of 0.00125 inch thickness (+ 20 percent tolerance). The bag shall be formed with heat sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag. The final clousure of the bag shall be heat sealed with the heat seal made as close as possible to the open end. Prior to or during the closure operation, excess air within the bag shall be expelled.

5.1.1.2 Intermediate packing. Ten lanyards of one class only, unit packed as specified in 5.1.1.1, shall be placed in a set-up paperboard box conforming to type I, variety 1, class A, style 4 of PPP-B-676; or a folding paperboard box conforming to variety 1, style III, type G, class i of PPP-B-566. Outside dimensions of the paperboard box shall be 8-1/2 inches in length, 5 inches in width, and 2-1/4 inches in depth. Box closure shall be secured with 2 inch minimum width gummed paper tape applied at the center of the length seam and completely circle the box in the girth dimension.

5.1.2 Commercial. Lanyards shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. One hundred lanyards of one class only, preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to grade V2s, style RSC of PPP-B-636. Level A intermediate packs shall be packed on end five in length, two in width, and one in depth within a shipping container. Inside dimensions of shipping container shall approximate 11-3/4 inches in length, 10-1/4 inches in width, and 8-3/4 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with Method III, waterproofed in accordance with Method V and reinforced as specified in the appendix PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.2 Level B packing. One hundred lanyards of one class only, preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A intermediate packs shall be packed on end five in length, two in width, and one in depth within a shipping container. Inside dimensions of shipping containers shall approximate 11-3/4 inches in length, 10-1/4 inches in width, and 8-3/4 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.5.

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5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be grade V3c, V3s or V4s, fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial packing. Lanyards, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), lanyards, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding mean K and L or film bonding means O or P. Pallet patterns shall be 104 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special; marking required by the contract or purchase order, intermediate packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951 as applicable.

5.4.1 Polyethylene unit packs. Polyethylene bagged unit packs shall have the required information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. The bag or label shall indicate the following information:

STOCK NUMBER
NOMENCLATURE
CLASS
QUANTITY

6. NOTES

6.1 Intended use. The pistol lanyard is intended for use by the Military Police of the Department of Defense. Class 1 is used by the Army and Air Force and class 2 is used by the Navy and Marine Corps.

6.2 Ordering data. Acquisition documents should specify this following:

a. Title, number, and date of this documents
b. Class required.
c. When a first article is required (see 3.1, 4.3, and 6.5)
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
f. When palletization is required (see 5.3).

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6.3 Samples. For access to a standard shade sample of nylon cover yarn, address the contracting activity issuing the invitation for bids.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

6.5 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions or FAR. 209. The first article should be a preproduction sample consisting of one completed lanyard. The contracting officer should include specific instructions in all documents, regarding arrangements for inspection, test and approval of the first article.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Preparing activity:

Army - GL

Project No. 8465-0912

Review activities:

Air Force - 82
DLA - CT

User activities:

Navy - MC
Air Force - 45

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-10028F	Lanyard, Individual Equipment Carrying

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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